TABLE I

CSO CONTACT GUIDANCE: SEVEN OBJECTS, 5 PAIRED TRIALS PER OBJECT AND MODEL. TIMES ARE MEAN ± STD; EXECUTION TIME IS SUCCESS-ONLY WALL TIME, AND PLANNING TIME COVERS ALL ROUNDS.

Metric	Contact Model			Solvers			
	SCM	MuJoCo QP	LCP	SQP	NLP	CMA-ES	CEM
Success Rate	45.7%	40.0%	20.0%	–	–	–	–
Execution Time (s)	12.06 ± 11.67	138.75 ± 165.29	46.51 ± 26.84	–	–	–	–
Planning Time (ms)	17.96 ± 4.89	123.45 ± 39.15	117.03 ± 29.94	–	–	–	–